

RESPONSE TO REQUEST FOR INFORMATION ON THE DEVELOPMENT OF AN ARTIFICIAL INTELLIGENCE (AI) ACTION PLAN

AI is transforming education, creating new opportunities to improve teaching and learning. However, AI integration requires a thoughtful approach that balances innovation with equity, transparency and accountability. By using AI responsibly, we can empower educators, personalize student learning, and ensure technology helps close, not widen, educational gaps. Achieving this vision will require sustained collaboration among students, parents, educators, policymakers, researchers, and developers. It will also require thoughtful design practices, robust testing, and reflection on the social and ethical implications of AI systems to ensure tools respond to students' strengths, needs, and cultural contexts. The following recommendations identify key actions to ensure AI supports safe, effective, and equitable learning environments.

RECOMMENDATIONS FOR AI IN EDUCATION

1. Research & Development: Driving AI Innovation for Learning

AI R&D must prioritize transformative, scalable, and accelerated educational innovations.

- **Expand Public-Private AI Research Collaborations** to ensure AI-driven solutions address unmet student learning needs while reinforcing, not replacing, educators' roles.
- **Establish AI Testbeds in Education** to evaluate real-world applications of AI in learning, balancing rapid advancements with clear, evidence-based student impact.
- **Advance AI for Adaptive Learning and Workforce Readiness** by supporting research in competency-based models, AI-assisted skills training, and scalable AI adoption pathways.
- **Incorporate Equity Audits in R&D Processes** to evaluate AI models for bias, ensuring systems are trained on representative data and tested for fairness across demographic groups.

2. K-12 (Primary and Secondary) AI Engagement: Empowering Schools, Educators, & Parents

K-12 leaders need clear guidance and practical resources to integrate AI responsibly and protect students.

- **Develop Research-based AI Implementation Guidelines** to help school leaders assess and deploy AI responsibly, ensuring rapid adoption does not outpace demonstrated effectiveness.
- **Strengthen AI Professional Development** to equip teachers with the skills to integrate AI while retaining instructional oversight.

- **Ensure AI Tools Align with Curricular Goals** to support teacher-student interactions and reinforce high-quality instructional models.
- **Promote Teacher-in-the-Loop Models** that prioritize teacher agency in AI decision-making and prevent over-reliance on automated systems.
- **Provide Schools with Tools to Evaluate Bias in AI Systems** by developing practical resources that help educators identify discriminatory outcomes in AI-driven platforms.

3. Postsecondary (Tertiary) AI Engagement: Preparing a Future-Ready Workforce

Postsecondary institutions play a vital role in advancing AI innovation and preparing students for emerging careers.

- **Strengthen Professional Development** for faculty and staff to support AI integration in instruction, assessment, and student supports.
- **Rigorously Test and Evaluate** AI tools to ensure alignment with educational goals and student needs.
- **Expand AI Workforce Pipelines** through industry partnerships that provide hands-on AI training and align with evolving job market demands.
- **Create Regional AI Learning Ecosystems** that foster collaboration across colleges, universities, and industry to prepare students for AI-driven career pathways.
- **Encourage Postsecondary Institutions to Adopt Practices that Identify and Mitigate Bias** in AI-enhanced admissions, assessment, and student support systems.

4. AI Literacy: Building Foundational Understanding

Equipping educators, students, and families with AI literacy skills is critical for informed engagement and responsible decision-making.

- **Provide Targeted AI Training for Educators** to ensure they can evaluate and apply AI-driven tools effectively.
- **Promote AI literacy as a Core Element of Workforce Readiness** by identifying and promoting STEM and non-STEM apprenticeships, technical certifications, and vocational programs to encourage digital readiness, innovation, and economic growth.
- **Empower Families and Communities** by establishing school-community partnerships and hands-on learning programs to ensure family engagement and community-driven solutions that strengthen the local understanding of AI literacy.
- **Advance Equity-Centered AI Literacy Programs** by designing materials accessible to marginalized groups, students with disabilities, and multilingual learners.

- **Provide Guidance on Algorithmic Bias for Educators and Families** to support informed engagement with AI systems and empower stakeholders to identify and challenge potentially discriminatory outcomes.

5. Industry Engagement: Ensuring AI Tools Support Education

Developers play a pivotal role in ensuring AI systems are safe, effective, and designed to improve learning outcomes before they are adopted in educational settings across all levels.

- **Establish AI Standards for Education Technology** to ensure transparency in AI functionality, data collection, and decision-making.
- **Encourage Investment in AI Safety & Effectiveness** by advocating for evidence-based standards that maintain educational integrity.
- **Develop AI Vendor Certification Programs** to ensure AI tools meet safety, privacy, and efficacy standards before school deployment.
- **Promote AI Co-Design Models** that encourage developers to actively engage educators, students, and parents during the design and testing phases to improve usability and relevance.
- **Encourage Developers to Engage Diverse Stakeholders in Design Phases** to ensure AI systems reflect a broad range of cultural, linguistic, and educational needs.

6. International Collaboration: Advancing Innovation and Equity in AI for Education

Global partnerships are essential to ensuring AI advances innovation, equity, and economic opportunity. By engaging in multilateral forums, the U.S. can shape global AI policies, expand opportunities for U.S. industry, and support educational equity worldwide.

- **Develop Collaborative AI Learning Networks** that connect education leaders across economies to exchange strategies for AI integration, address shared challenges, and adapt solutions to diverse educational contexts.
- **Create Joint AI Governance Frameworks** with international partners to promote transparency, mitigate bias, and ensure AI systems are inclusive, accessible, and aligned with democratic values.
- **Support Cross-Border AI Workforce Readiness Initiatives** by aligning AI literacy frameworks, sharing career development strategies, and ensuring students are prepared for global job markets in AI-driven industries.

CONCLUSION

AI's role in education is not just about innovation — it's about shaping the future of learning in ways that expand opportunity and uphold core educational values. By approaching AI with

intention, we can unlock its potential to drive equity, empower educators, and prepare students to thrive in an increasingly complex world. The true measure of success will be our ability to develop and apply AI in ways that reflect our highest aspirations for education — fostering curiosity, creativity, and human connection while safeguarding the rights and well-being of every learner.

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ENDNOTES

1. U.S. Department of Education, [*Artificial Intelligence and the Future of Teaching and Learning: Insights and Recommendations*](#), Washington, DC, 2023.
2. U.S. Department of Education, [*Designing for Education with Artificial Intelligence: An Essential Guide for Developers*](#), Washington, DC, 2024.
3. U.S. Department of Education, [*Empowering Education Leaders: A Toolkit for Safe, Ethical, and Equitable AI Integration*](#), Washington, DC, 2024.
4. U.S. Department of Education, [*Navigating Artificial Intelligence in Postsecondary Education: Building Capacity for the Road Ahead*](#), Washington, DC, 2025.